AIFS-DATASET for Few-Shot Aerial Image Scene Classification

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***Abstract*— Few-shot learning (FSL), which aims to rapidly recognize unseen categories with limited samples, has attracted wide attention in aerial image scene classification. However, the existing methods generally train and evaluate the model within a dataset, and changing the dataset requires retraining and evaluation, which only realizes the generalization of intra-dataset. Considering meta-learning, this brings in a natural assumption: FSL should learn meta-knowledge from cross-domain heterogeneous tasks and then can generalize to new data distributions (e.g., datasets) with few samples. To this end, we propose a new benchmark, dubbed aerial image few-shot dataset (AIFS-DATASET), which is composed of diverse datasets and can provide more realistic heterogeneous task distributions. On AIFS-DATASET, we use many heterogeneous tasks, across multi-domains without any aerial image category, to train the model, achieving "see more." Then we transfer the learned knowledge to new tasks in aerial images to evaluate the generalization performance of the model, thus acquiring a "well-informed" few-shot aerial image scene classification model. Moreover, the challenges of inter-class similarity and intra-class discrepancy in aerial images still exist. We also develop a dual constrained distance metric learning (DC-DML) framework to deal with the variable learning tasks adaptively and to achieve compact data distribution within a class and clear distribution gaps between classes from the perspective of metric learning. DC-DML mainly uses a task-adapted feature extractor while devising a novel distance metric with a cross-class bias penalty. By conducting experiments on AIFS-DATASET, we observed that DC-DML outperforms the current prevailing FSL approaches by a large margin.**

***Index Terms*— Aerial images, few-shot learning (FSL), metric learning, scene classification.**

I. Introduction

DEEP learning has made tremendous advances in aerial image scene classification [1]–[7]. What makes these methods so remarkable, however, is a high dependency on large amounts of annotated data. Typically, collecting and annotating gigantic aerial data is time-consuming and expensive, especially for complex aerial images [8]. Besides, such learned deep learning models excel at processing tasks within the scope defined by training classes but have limited model performance on novel classes. Because of the foregoing facts, few-shot learning (FSL) [9]–[16] emerges as the times require, which aims at endowing machines with quick learning and reasoning ability of humans.

Due to the promising learning ability of FSL, a growing number of researchers apply FSL to the field of remote sensing images. Despite notable successes, the existing FSL works normally use a benchmark in which both the training set and test set are derived from the same dataset, and the trained model can be adapted to the test (or novel) classes with only minor or no adjustments at all [17]–[19]. Besides, these methods need to be retrained and evaluated when performing experiments on another dataset. In doing so, these methods ignore the importance of out-of-dataset evaluation while seeking to realize the generalization of intra-dataset.

Currently, a prevailing direction to address an FSL challenge is to train the model under the meta-learning paradigm with an episodic training strategy [20]–[25]. Specifically, given several training tasks, the model learns prior knowledge across these tasks and then transfers the knowledge to quickly recognize novel classes not seen in the training stage, with little labeled samples. Such learning tasks among the existing methods are homogeneous task distribution. The numbers of categories and samples between each task are the same. Whereas real-life learning tasks tend to be heterogeneous, that is, the numbers of categories and samples among multiple heterogeneous tasks are not uniform. Therefore, this brings in a natural assumption: FSL should learn meta-knowledge from cross-domain heterogeneous tasks and then can generalize to new data distributions (e.g., datasets) with few samples. This assumption always holds in practice. Thus, we advocate that a more realistic and challenging benchmark needs to be constructed for FSL in the aerial image area.

To this end, in this article, inspired by Meta-dataset [26], we construct a new benchmark, called aerial image few-shot dataset (AIFS-DATASET), which is composed of multiple databases with diverse data distribution. In task creation of AIFS-DATASET, the number of categories and samples among each task is variable, aiming to achieve heterogeneous task distribution. More specifically, on AIFS-DATASET, we use many cross-domain heterogeneous tasks without any aerial image category, to train the model, achieving "see more." Then we transfer the learned knowledge to new tasks in aerial images to evaluate the generalization performance of the model, thus acquiring a "well-informed" few-shot aerial image scene classification model. To sum up, the purpose of constructing AIFS-DATASET is to more rigorously verify the

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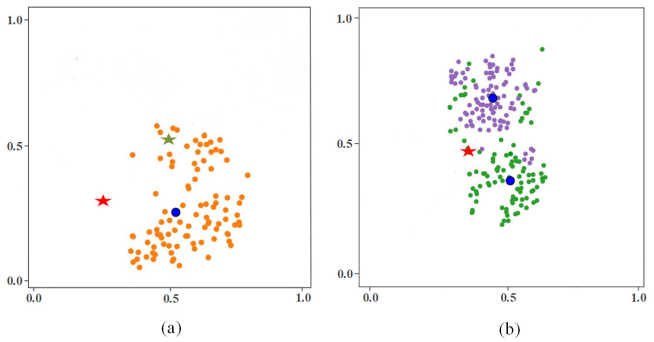


Fig. 1. Influence of sample feature distribution on distance metric. (a) Intra-class differences lead to discretization of intra-class sample distribution. (b) Inter-class similarity results in blurring of interval between classes.

generalization ability of the FSL model in aerial image scene classification, and also to get closer to the practical application situation. With the increase in task heterogeneity, the demand for the adaptive ability of the model is correspondingly higher. Therefore, we leverage the feature extractor in the conditional neural adaptive processes (CNAPs) [27] approach to adaptively extract sample features.

In addition, due to the problems of inter-class similarity and intra-class discrepancy in aerial images, it is necessary to consider the influence of sample features' distribution when measuring the similarity between sample features. First, the discrepancy between different samples of the same category results in the discretization of the sample's distribution within the class and interferes with similarity calculation. As shown in Fig. 1(a), red and green stars are equidistant from the center of distribution. However, it can be clearly seen from Fig. 1(a) that the green star should be the point within distribution and the red star is the point outside distribution. Accordingly, it is reasonable that the distance between the green star and the center of distribution should be closer. Second, overlap may exist between sample distribution of different categories, which blurs the category boundaries. As shown in Fig. 1(b), the red star is at the same distance from the center of both distributions, but we can see from Fig. 1(b) that the red star belongs to green distribution. To circumvent these problems, we propose a new metric learning, referred to as dual constrained distance metric learning (DC-DML). Specifically, DC-DML estimates two different matrices by leveraging current task data distribution and combines the metric learning and a cross-class bias penalty term, to implement the goal of narrowing inter-class similarity and intra-class discrepancy, thereby improving few-shot scene classification accuracy in aerial images.

In summary, this work makes the following contributions.

1) We construct a new realistic benchmark, dubbed AIFS-DATASET, for FSL of aerial image scene classification.
2) We propose a DC-DML to mitigate the problems of inter-class similarity and intra-class discrepancy in aerial images.
3) We perform the experimental evaluation of popular approaches and ablation study of our approach. The experimental results demonstrate the effectiveness of our DC-DML.

The rest of this article is organized as follows. Section II gives a brief review of few-shot works in remote sensing images and metric learning. In Section III, we describe the new benchmark AIFS-DATASET and DC-DML in detail. Section IV presents comprehensive experimental results on AIFS-DATASET. Finally, Section V provides the conclusion of this article.

II. Related Works

In this section, we will review related works in two aspects: few-shot works in remote sensing image scene classification and metric learning.

A. FSL of Remote Sensing Scene Classification

With the rise of deep learning, a large number of remote sensing image classification models have emerged. Nogueira *et al.* [28] used three training strategies to learn convolutional neural networks (ConvNets) in the remote sensing scene classification task and concluded that the fine-tuned ConvNets with support vector machine (SVM) can achieve excellent performance. Lu *et al.* [29] proposed a shallow weighted deconvolution network with a spatial pyramid model (SPM) and SVM to perform unsupervised representation learning for remote sensing scene classification. Cheng *et al.* [30] proposed a large-scale benchmark, called "NWPU-RESISC45," which includes 45 scene categories with 700 samples per category. Hong *et al.* [31] developed a general multimodal deep learning (MDL) model to overcome the limitations of remote sensing information diversity. Wan *et al.* [32] proposed multiscale dynamic GCN (MDGCN) to accurately classify hyperspectral images (HSIs). A new minibatch GCN (miniGCN) [33] is proposed to improve HSI classification by integrating CNN and graph convolutional network (GCN). Hang *et al.* [34] developed an attention-aided CNN model for HSIs' classification.

However, restricted by the quantity and quality of manually annotated semantic labels, the performance of the above scene classification models may be limited. Fortunately, a new research direction has emerged, namely, FSL, which can identify new categories with a small number of labeled samples to enhance the generalization performance of the model [35]. FSL has also been widely studied in remote sensing image classification. Rostami *et al.* [36] developed a classification algorithm to alleviate the scarcity of synthetic aperture radar (SAR) annotation samples by transferring knowledge from the electro-optical (EO) domain to the SAR domain. Liu *et al.* [37] designed a deep 3-D convolutional neural network for few-shot HSI classification. Li *et al.* [38] proposed a DLA-MatchNet for boosting a few-shot remote sensing image scene classification. Li *et al.* [39] proposed an RS-MetaNet and Balance Loss with task-level training manner to solve the easy overfitting issue of FSL in remote sensing scene classification. Despite notable successes, all these models leverage the benchmark with a single data distribution or dataset to train and test the generalization performance of the model, which obviously cannot ensure the reliability of generalization. Combined with real-life learning situation

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and meta-learning, FSL should learn meta-knowledge from cross-domain heterogeneous tasks and then can generalize to new data distributions (e.g., datasets) with few samples. However, the maximum number of remote sensing categories is 45 on the existing publicly available remote sensing datasets, and the categories among these datasets are mostly repeated, so the diversity of cross-domain data distribution and the generation of heterogeneous meta-tasks cannot be guaranteed. Therefore, we need to train the model on the dataset of rich and diverse categories, and then verify the FSL ability of the model for remote sensing images. In brief, a more realistic and challenging benchmark is still lacking.

B. Metric Learning

Currently, metric learning [40] is also used in FSL. Vinyals *et al.* [41] integrated cosine distance and augment neural networks with external memories to perform one-shot learning on Omniglot and ImageNet datasets. Snell *et al.* [42] proposed a prototypical network (PN), which can learn a metric space using squared Euclidean distance and ConvNets, for FSL. Sung *et al.* [43] developed a relation network (RN) to learn a transferrable deep metric for few-shot image classification. Lichtenstein *et al.* [44] proposed a task-adaptive feature sub-space learning (TAFSSL) framework, which uses the nearest neighbor (NN) for few-shot classification. For now, however, effective metric learning in the few-shot aerial image scene classification task is absent.

III. Proposed Benchmark and Method

A. Aerial Image Few-Shot Dataset (AIFS-DATASET)

The main role of AIFS-DATASET is to provide a rigorous and realistic benchmark for evaluating few-shot scene classification in aerial images. Next, we will describe our new benchmark from dataset structure and task creation in detail.

1) Dataset Structure: AIFS-DATASET is composed of multiple datasets with varied types, which breaks the drawback of the generalization of intra-dataset. Specifically, AIFS-DATASET mainly contains the following ten datasets: CUB-200-2011 [45], Describable Textures [46], Fungi [47], VGG Flower [48], Traffic Signs [49], CIFAR100 [50], NWPU-RESISC45 [30], AID [51], UC Images [52], and WHU-RS19 [53]. The reasons for choosing the first six datasets are: 1) these datasets are publicly available; 2) these datasets are also used for image classification, and almost all of them have the challenge of large intra-class diversity and small inter-class difference, which happens to be the challenge of remote sensing image scene classification; 3) there is not much overlap between the categories in these natural image datasets and those in remote sensing datasets; and 4) the differences and diversity between these datasets are also large enough, as shown in Fig. 2. Due to task requirements of few-shot aerial image scene classification, we choose four aerial image datasets of NWPU-RESISC45, AID, UC Images, and WHU-RS19 for evaluation. For the remaining datasets, each dataset is divided into training category set $\mathcal{C}_{\text{train}}$, validation category set $\mathcal{C}_{\text{val}}$, and test category set $\mathcal{C}_{\text{test}}$ in a ratio of 70%/15%/15%, and these category sets are not intersected with each other. Moreover, to avoid the issue of category overlap between the training category set and aerial image datasets (such as aircraft, car, and bridge), we separate these overlapping categories from the training category set and add these categories to the validation category set or test category set.

2) Task Creation: In FSL of aerial image scene classification, the training set $\mathcal{D}_{\text{tr}} = \{(x_i, y_i)\}_{i=1}^{N}$ contains a number of categories, each of which includes many samples. N is the number of samples in the training set $\mathcal{D}_{\text{tr}}$, x_i represents the ith sample in $\mathcal{D}_{\text{tr}}$, and y_i is the category label of x_i. During the meta-training stage, C categories are randomly selected from the training set $\mathcal{D}_{\text{tr}}$, and K samples are selected from each category, with a total sample number of $C \times K$, forming a support set $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^{C\times K}$. A part of samples from the remaining samples are selected as a query set $\mathcal{Q} = \{(x_i^*, y_i^*)\}_{i=1}^{m}$. This is called a C-way K-shot, and a support set $\mathcal{S}$ and a query set $\mathcal{Q}$ constitute a meta-task $\mathcal{T}$. The primary purpose of the meta-training stage is to learn a meta-learner in an episodic fashion. Different meta-tasks are sampled, which means different category combinations. This enables the model to learn sample similarity and important features, and to synthesize the common parts of different meta tasks, thereby new tasks can be learned quickly in the testing stage. To obtain heterogeneous task distribution, it is necessary to ensure the variable shots and ways within each task. To this end, we first sample the categories from the training set $\mathcal{D}_{\text{tr}}$ according to the interval [5, MAXCLASSES], to realize the variability of "way." Then, we sample support and query samples from the selected categories. Thereinto, the support samples are also sampled from each category according to the interval [1, MAXSUPPORT], ensuring the variability of "shot." The number of query samples is class-balanced for facilitating evaluation.

B. Dual Constrained Distance Metric Learning (DC-DML)

We use a CNAPs' approach to make the model adaptive to heterogeneous tasks. Considering the computational cost of CNAPs and the issues of inter-class similarity and intra-class diversity in aerial images, we propose a DC-DML method to improve the CNAPs' approach. The architecture of DC-DML is illustrated in Fig. 3. Since our DC-DML method is derived from the CNAPs' approach, we first introduce the CNAPs' approach in detail before describing DC-DML.

1) Conditional Neural Adaptive Processes (CNAPs): The CNAPs' approach is an improvement on a conditional neural processes (CNPs) [54] approach. Given a task $\mathcal{T}$, it contains a support set $\mathcal{S} = \{(x_i, y_i)\}_{i=1}^{C\times K}$ and a query set $\mathcal{Q} = \{(x_i^*, y_i^*)\}_{i=1}^{m}$. The model uses the support set $\mathcal{S}$ to predict the label y^* of the sample x^* in the query set $\mathcal{Q}$ through the following:

$$p\left(y^*|x^*, \vartheta, \mathcal{S}\right) = p\left(y^*|x^*, \vartheta, \Phi^{\mathcal{T}} = \Phi_\phi(\mathcal{S})\right). \tag{1}$$

Parameter $\Phi^{\mathcal{T}}$ is local task-specific, which assists the global parameter ϑ to realize the task-adaptive classification learning of the model. $\Phi^{\mathcal{T}}$ is obtained from the adaptive network Φ_ϕ and the support set $\mathcal{S}$, and ϕ is a global parameter. CNAPs mainly improve the adaptive network Φ_ϕ by designing

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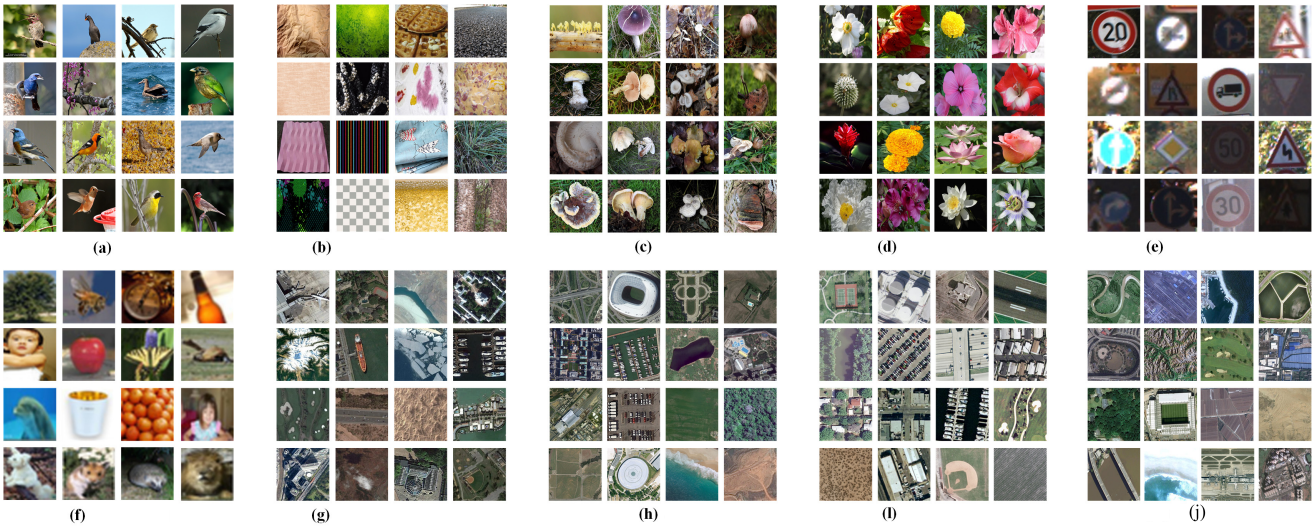


Fig. 2. Various datasets contained in AIFS-DATASET. (a) CUB. (b) Describable Textures. (c) Fungi. (d) VGG Flower. (e) Traffic Signs. (f) CIFAR100. (g) NWPU-RESISC45. (h) AID. (i) UC Images. (j) WHU-RS19.

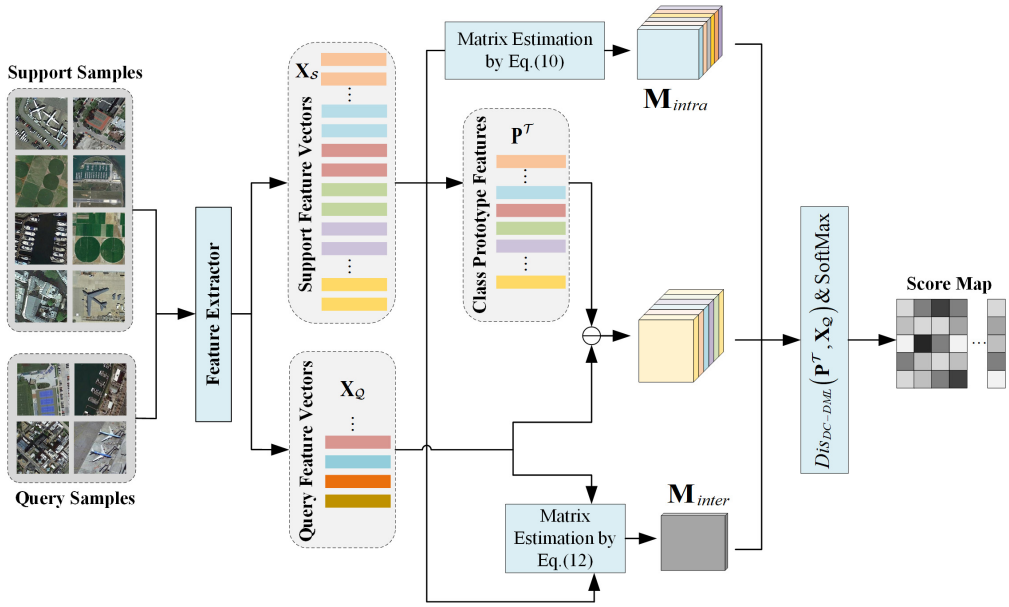


Fig. 3. Overall framework of DC-DML.

a task-adapted feature extractor and a task-adapted classifier to improve the flexibility of the model. Specifically, for the task-adapted feature extractor (see Fig. 3), a feature-wise linear modulation (FiLM) layer [55] is added to the ResNet18 network to linearly adjust feature maps, allowing the feature extractor to dynamically adapt to few-shot multi-task learning. After passing through the FiLM layer, the feature map f_i of layer i is updated as

$$\mathrm{FiLM}\left(f_i; \gamma_i^{\mathcal{T}}, \beta_i^{\mathcal{T}}\right) = \gamma_i^{\mathcal{T}} f_i + \beta_i^{\mathcal{T}}. \tag{2}$$

The role of the parameter $\Phi_f^{\mathcal{T}} = \{\gamma^{\mathcal{T}}, \beta^{\mathcal{T}}\}$ is the same as that of the parameter $\Phi^{\mathcal{T}}$ in (1). $\Phi_f^{\mathcal{T}}$ is obtained using the network $\Phi_{\phi^f}(\cdot; \phi^f, \vartheta)$, and ϕ^f denotes the parameter of Φ_{ϕ^f}. The input at layer i of the network Φ_{ϕ^f} is $z_i = (z_G^{\mathcal{T}}, z_{AR}^i)$. $z_G^{\mathcal{T}}$ is obtained from a task set encoder g. z_{AR}^i is obtained by adaptive activation of the previous convolutional block in feature extractor. Our DC-DML method uses this task-adapted feature extractor. For the task-adapted classifier, the data features and a DenseResNet Φ_{ϕ^w} [56] are used to determine the classifier weight $\Phi_w^{\mathcal{T}} \in \mathbb{R}^{d \times C^{\mathcal{T}}}$, ϕ^w denotes the parameter of Φ_{ϕ^w}, d is the feature dimension, and $C^{\mathcal{T}}$ is the number of categories within a task $\mathcal{T}$. Each column of the weight matrix $\Phi_w^{\mathcal{T}} = [\Phi_{\phi^w}(\mathcal{S}_1^{\mathcal{T}}), \ldots, \Phi_{\phi^w}(\mathcal{S}_c^{\mathcal{T}}), \ldots, \Phi_{\phi^w}(\mathcal{S}_{C^{\mathcal{T}}}^{\mathcal{T}})]$ is independently generated by the features of each category in the support set $\mathcal{S}$. $\mathcal{S}_c^{\mathcal{T}}$ refers to the sample set of c_{th} category in the support set $\mathcal{S}$ within a task $\mathcal{T}$. For example, the c_{th} column of the weight matrix $\Phi_w^{\mathcal{T}}$ can be computed as

$$\begin{aligned}&\Phi_{\phi^w}\left(\mathcal{S}_c^{\mathcal{T}}; \phi^w, \phi^f, \vartheta\right)\\ &\quad = \Phi_{\phi^w}\left(\left\{\mathcal{F}\left(\Phi_{\phi^f}(x_j; \phi^f, \vartheta)\right) | x_j \in \mathcal{S}_c^{\mathcal{T}}, y_j = c\right\}; \phi^w\right)\end{aligned} \tag{3}$$

where $\mathcal{F}$ denotes the task-adapted feature extractor. The global parameter ϑ can be acquired by general classification training. For CNAPs, $\Phi^{\mathcal{T}}$ in (1) can be expressed as $\Phi^{\mathcal{T}} = \{\Phi_f^{\mathcal{T}}, \Phi_w^{\mathcal{T}}\}$; likewise, the adaptive network Φ_ϕ in (1) can also be expressed as $\Phi_\phi = \{\Phi_{\phi^w}, \Phi_{\phi^f}\}$. Parameter $\Phi^{\mathcal{T}}$ can be obtained through episodic training, and the global parameter ϑ is fixed during

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episodic training

$$\mathcal{L}(\Phi^{\mathcal{T}}) = \frac{1}{MT}\sum \log p\left(y_i^{*\mathcal{T}} | x_i^{*\mathcal{T}}, \mathcal{F}(\mathcal{S}^{\mathcal{T}}; \Phi^{\mathcal{T}})\right). \quad (4)$$

Equation (4) is the loss function with respect to parameter $\Phi^{\mathcal{T}}$. $\mathcal{S}^{\mathcal{T}}$ refers to the support set in the task $\mathcal{T}$, and $x_i^{*\mathcal{T}}$ is the i_{th} query sample of the task $\mathcal{T}$. $y_i^{*\mathcal{T}}$ is the label of $x_i^{*\mathcal{T}}$. M is the total number of query samples, and T is the total number of meta-tasks.

2) Dual Constrained Distance Metric Learning (DC-DML): Generally, the distance metric in FSL is to calculate the distance between the query sample feature $\mathcal{F}(x^*; \Phi_f^{\mathcal{T}}, \Phi_w^{\mathcal{T}})$ and each class prototype feature of the support set $\mathcal{S}$ within a task $\mathcal{T}$, and then apply SoftMax to these distances to get the category probability distribution of the query sample x^* in task $\mathcal{T}$

$$p\left(y^* = c | x^*\right) = \frac{\exp\left(-d\left(\mathcal{F}\left(x^*; \Phi_f^{\mathcal{T}}, \Phi_w^{\mathcal{T}}\right), \mathbf{P}_c\right)\right)}{\sum_k \exp\left(-d\left(\mathcal{F}\left(x^*; \Phi_f^{\mathcal{T}}, \Phi_w^{\mathcal{T}}\right), \mathbf{P}_k\right)\right)} \quad (5)$$

$$\mathbf{P}_c = \frac{1}{|\mathcal{S}_c|}\sum_{(x_i, y_i)\in\mathcal{S}_c} \mathcal{F}\left(x_i; \Phi_f^{\mathcal{T}}, \Phi_w^{\mathcal{T}}\right) \quad (6)$$

$$d(x_1, x_2) = \sqrt{(x_1 - x_2)^T (x_1 - x_2)}. \quad (7)$$

For convenience, we use $\mathcal{F}(x)$ for $\mathcal{F}(x; \Phi_f^{\mathcal{T}}, \Phi_w^{\mathcal{T}})$ in the following content. $\mathbf{P}_c$ denotes the class prototype feature of the c_{th} category. $\mathcal{S}_c$ refers to the support samples of the c_{th} category in the task $\mathcal{T}$. However, due to the problems of intra-class difference and inter-class similarity in aerial images, the influence of sample features distribution should be considered when calculating the distance between query sample feature and class prototype feature. To this end, we propose a DC-DML method to improve few-shot scene classification in aerial images. Specifically, given a meta-task $\mathcal{T}$, we use the task-adapted feature extractor of CNAPs to extract the query sample feature $\mathbf{X}_{\mathcal{Q}} \in \mathbb{R}^{N_{\mathcal{Q}}\times d}$ and the support sample feature $\mathbf{X}_{\mathcal{S}} \in \mathbb{R}^{N_{\mathcal{S}}\times d}$. d is the number of feature dimension. $N_{\mathcal{S}}$ and $N_{\mathcal{Q}}$ are the number of samples of the support set $\mathcal{S}$ and query set $\mathcal{Q}$, respectively. The distance matrix $\mathbf{D}$ between the query sample feature $\mathbf{X}_{\mathcal{Q}}$ and the class prototype feature $\mathbf{P}^{\mathcal{T}} \in \mathbb{R}^{C^{\mathcal{T}}\times d}$ in the support set $\mathcal{S}$ can be expressed as

$$\mathbf{D} = \left(\mathbf{P}^{\mathcal{T}} - \mathbf{X}_{\mathcal{Q}}\right) \in \mathbb{R}^{C^{\mathcal{T}}\times N_{\mathcal{Q}}\times d} \quad (8)$$

where $C^{\mathcal{T}}$ is the number of categories in task $\mathcal{T}$. The c_{th} row in the class prototype feature $\mathbf{P}^{\mathcal{T}}$ represents the class prototype feature of the c_{th} category in the support set $\mathcal{S}$. First, to eliminate the intra-class differences, we use a matrix $\mathbf{M}_{\text{intra}} \in \mathbb{R}^{C^{\mathcal{T}}\times d\times d}$ to perform a linear transformation on query sample feature $\mathbf{X}_{\mathcal{Q}}$ and support set sample feature $\mathbf{X}_{\mathcal{S}}$, and then the distance matrix can be rewritten as

$$\hat{\mathbf{D}} = \mathbf{M}_{\text{intra}} \times \left(\mathbf{P}^{\mathcal{T}} - \mathbf{X}_{\mathcal{Q}}\right) \in \mathbb{R}^{C^{\mathcal{T}}\times N_{\mathcal{Q}}\times d}. \quad (9)$$

As shown in Fig. 4(a), we perform this linear transformation within each class. Matrix $\mathbf{M}_{\text{intra}}$ is estimated from the support samples of task $\mathcal{T}$, and the c_{th} dimension of $\mathbf{M}_{\text{intra}} \in \mathbb{R}^{C^{\mathcal{T}}\times d\times d}$

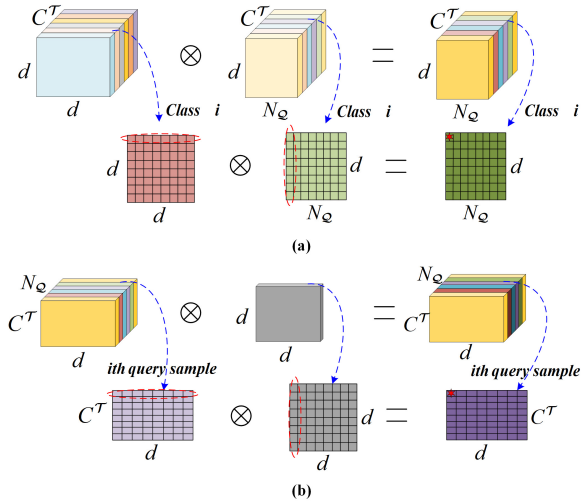


Fig. 4. Illustration of our distance metric learning operation. (a) Performing a linear transformation within each class. (b) Implementing inter-class constraints.

in the range $1 \sim C^{\mathcal{T}}$ can be represented as follows:

$$\mathbf{M}_{\text{intra}}^c = \mathbf{I} + \frac{1}{N_{\mathcal{S}}^c - 1} \times \sum_{(x_i, y_i)\in\mathcal{S}_c} (\mathcal{F}(x_i) - \mathbf{P}_c)(\mathcal{F}(x_i) - \mathbf{P}_c)^T \in \mathbb{R}^{1\times d\times d} \quad (10)$$

where $\mathcal{S}_c$ denotes the sample set of class c in the support set $\mathcal{S}$. $N_{\mathcal{S}}^c$ is the number of samples in $\mathcal{S}_c$. Second, to enhance the separability between classes, we further use a matrix $\mathbf{M}_{\text{inter}} \in \mathbb{R}^{1\times d\times d}$ to constrain feature metric calculation, as shown in Fig. 4(b)

$$\tilde{\mathbf{D}} = \hat{\mathbf{D}} \times \mathbf{M}_{\text{inter}} = \mathbf{M}_{\text{intra}} \times \left(\mathbf{P}^{\mathcal{T}} - \mathbf{X}_{\mathcal{Q}}\right) \times \mathbf{M}_{\text{inter}} \in \mathbb{R}^{N_{\mathcal{Q}}\times C^{\mathcal{T}}\times d}. \quad (11)$$

Matrix $\mathbf{M}_{\text{inter}}$ is estimated from all the samples in task $\mathcal{T}$

$$\mathbf{M}_{\text{inter}} = \frac{1}{N-1}\sum_{(x_i)\in\mathcal{T}} (\mathcal{F}(x_i) - \mu)(\mathcal{F}(x_i) - \mu)^T \in \mathbb{R}^{1\times d\times d}. \quad (12)$$

$N = N_{\mathcal{S}} + N_{\mathcal{Q}}$ is the number of samples in task $\mathcal{T}$. μ is the mean of all the samples in $\mathcal{T}$.

Our metric design is an improvement on the Euclidean distance. Assume that x_1 and x_2 are two samples in task $\mathcal{T}$, the Euclidean distance of x_1 and x_2 is

$$\text{Dis}_{\text{Euclidean}}(x_1, x_2) = \sqrt{(x_1 - x_2)^T (x_1 - x_2)}. \quad (13)$$

Assuming that $x_1 = Uz_1$ and $x_2 = Uz_2$, then $z_1 = t_1 V$ and $z_2 = t_2 V$, and the Euclidean distance of x_1 and x_2 can be rewritten as

$$\begin{aligned}\text{Dis}_{\text{Euclidean}}(x_1, x_2) &= \sqrt{(x_1 - x_2)^T (x_1 - x_2)} \\ &= \sqrt{(Uz_1 - Uz_2)^T (Uz_1 - Uz_2)} \\ &= \sqrt{(Ut_1V - Ut_2V)^T (Ut_1V - Ut_2V)} \\ &= \sqrt{V^T (t_1 - t_2)^T U^T U (t_1 - t_2) V}. \end{aligned} \quad (14)$$

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In our DC-DML, the distance formula between query sample feature $\mathbf{X}_{\mathcal{Q}}$ and class prototype feature $\mathbf{P}^{\mathcal{T}} \in \mathbb{R}^{C^{\mathcal{T}} \times d}$ in the support set $\mathcal{S}$ is

$$\begin{aligned}
&\mathrm{Dis}_{\text{DC-DML}}\left(\mathbf{P}^{\mathcal{T}}, \mathbf{X}_{\mathcal{Q}}\right) \\
&= \left\|\mathbf{M}_{\text{intra}}\left(\mathbf{P}^{\mathcal{T}} - \mathbf{X}_{\mathcal{Q}}\right)\mathbf{M}_{\text{inter}}\right\|_2^2 \\
&= \left(\mathbf{M}_{\text{intra}}\left(\mathbf{P}^{\mathcal{T}} - \mathbf{X}_{\mathcal{Q}}\right)\mathbf{M}_{\text{inter}}\right)^T\left(\mathbf{M}_{\text{intra}}\left(\mathbf{P}^{\mathcal{T}} - \mathbf{X}_{\mathcal{Q}}\right)\mathbf{M}_{\text{inter}}\right) \\
&= \mathbf{M}_{\text{inter}}^T\left(\mathbf{P}^{\mathcal{T}} - \mathbf{X}_{\mathcal{Q}}\right)^T\mathbf{M}_{\text{intra}}^T\mathbf{M}_{\text{intra}}\left(\mathbf{P}^{\mathcal{T}} - \mathbf{X}_{\mathcal{Q}}\right)\mathbf{M}_{\text{inter}}. \quad (15)
\end{aligned}$$

Hence, according to (14), when $\mathbf{M}_{\text{intra}} = \mathbf{I}$ and $\mathbf{M}_{\text{inter}} = \mathbf{I}$, our DC-DML distance is equivalent to the squared Euclidean, and $\mathbf{I}$ is an identity matrix.

Moreover, for task creation of AIFS-DATASET, the data distribution between tasks is different, for instance, the samples for task $\mathcal{T}_1$ are from CUB-200-2011, while the samples for task $\mathcal{T}_2$ may be sampled from Fungi. Since the samples of the support set $\mathcal{S}$ and query set $\mathcal{Q}$ in the task $\mathcal{T}$ are from the same dataset, the samples of the two sets are distributed in the same domain. We introduce a cross-class bias penalty into the objective function to realize compact distribution of samples within a task (or domain), so as to further improve the classification ability of the model. The overall objective function is as follows:

$$\begin{aligned}
\mathcal{L} &= \frac{1}{MT}\Big[\sum \log p\left(y_m^{*\mathcal{T}} | x_m^{*\mathcal{T}}, \mathcal{F}\left(\mathcal{S}^{\mathcal{T}}; \Phi_f^{\mathcal{T}}, \Phi_w^{\mathcal{T}}\right)\right) \\
&\qquad + \sum \left\|\mathbf{X}_{\mathcal{Q}^{\mathcal{T}}} - \mathbf{X}_{\mathcal{S}^{\mathcal{T}}}\right\|\Big] \\
p\left(y^* = c | x^*\right) &= \frac{\exp\left(-\mathrm{Dis}_{\text{DC-DML}}\left(\mathcal{F}\left(x^*; \Phi_f^{\mathcal{T}}, \Phi_w^{\mathcal{T}}\right), \mathbf{P}_c\right)\right)}{\sum_k \exp\left(-\mathrm{Dis}_{\text{DC-DML}}\left(\mathcal{F}\left(x^*; \Phi_f^{\mathcal{T}}, \Phi_w^{\mathcal{T}}\right), \mathbf{P}_k\right)\right)}. \quad (16)
\end{aligned}$$

IV. Experiments

In this section, we perform the experimental evaluation of popular approaches on AIFS-DATASET and ablation study of our DC-DML method.

A. Datasets

In AIFS-DATASET, in addition to aerial image datasets (i.e., NWPU-RESISC45, AID, UC Images, and WHU-RS19) used for evaluation, the remaining datasets are divided into training sets, verification sets, and test sets: 140 training classes, 30 verification classes, and 30 test classes for CUB-200-2011; 33 training classes, seven verification classes, and seven test classes for Describable Textures; 994 training classes, 200 verification classes, and 200 test classes for Fungi; 71 training classes, 15 verification classes, and 16 test classes for VGG Flower; 30 training classes, six verification classes, and seven test classes for Traffic Signs; 72 training classes, 13 verification classes, and 15 test classes for CIFAR100. In the above six datasets, except for the categories that overlap with the aerial image, the categories are used as validation or test classes, and the rest of the categories are randomly divided. We use the training sets of the six datasets (i.e., CUB-200-2011, Describable Textures, Fungi, VGG Flower, Traffic Signs, and CIFAR100) to train the model, and the test sets of these six datasets for the "in-domain" few-shot classification evaluation. The four aerial image datasets are used to evaluate the "out-of-domain" performance.

B. Implementation Details

Consistent with CNAPs, we also use a ResNet18 with FiLM layers as the feature extraction backbone, where ResNet18 is pretrained on the ImageNet dataset. Then, the parameters of ResNet18 are fixed, and the parameters of other layers (e.g., FiLM layers) are trained in an episodic training way. The structure of the feature adaptation extractor is the same as that of the feature extraction structure of CNAPs. We choose Adam as the optimizer with a learning rate of 0.0005. The training iteration is set to 100 000, and the evaluation results are averaged from 2000 episodes randomly sampled with 95% confidence interval. The number of categories included in each episode is randomly selected from the interval [5, MAXCLASSES], and the number of support samples for each category is randomly determined from the interval [1, MAXSUPPORT]. In our experiment, MAXCLASSES is set to 5/10/20, and MAX SUPPORT is set to 5/10/20.

C. Experimental Results and Comparisons

To verify the effectiveness of our method, we compare our DC-DML with several popular approaches, including k-NN [27], Finetune [57], MatchingNet [41], ProtoNet [42], fo-MAML [58], RelationNet [43], fo-Proto-MAML [26], CNAPs [27], and SimpleCNAPs [59]. Tables I–III show the experimental results of DC-DML and comparison approaches.

1) In-Domain Performance: We first report the in-domain results for DC-DML and comparison approaches. In-domain performance mainly refers to classification performance over six datasets, including CUB-200-2011, Describable Textures, Fungi, VGG Flower, Traffic Signs, and CIFAR100. From Table I, it can be seen that DC-DML obtains the best average classification accuracy of the in-domain at 58.8%, a 0.5% gain over RelationNet. DC-DML also exceeds CNAPs and SimpleCNAPs, which use the same task-adapted feature extractor as DC-DML, by 5.1% and 1.3%, respectively. DC-DML outperforms the comparison approaches on three out of six datasets in Table I. More specifically, on Describable Textures, Fungi, and VGG Flower, DC-DML achieves 4.6%, 0.8%, and 8.3% boost over the best comparison approaches, respectively. In short, the in-domain performance gains of Table I are considerable with 0.5%–8% margins. For Table II, on Describable Textures, VGG Flower, and CIFAR100, DC-DML surpasses the performance of the suboptimal approaches with 1.3%, 1.8%, and 0.9% margins, respectively. For the average accuracy of in-domain classification, although our DC-DML is 1.9% worse than CNAPs, overall, DC-DML achieves the highest classification accuracy in three of the six datasets. Similarly, in Table III, on Describable Textures and CIFAR100, DC-DML obtains 5.6% and 2% improvements over the suboptimal CNAPs, respectively. On VGG Flower, DC-DML obtains 76.9% of the classification accuracy, which is less than 0.1% to the best comparison approach. From

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TABLE I

FEW-SHOT CLASSIFICATION ACCURACY OF DC-DML COMPARED WITH THE PREVAILING APPROACHES. MAXCLASSES = 5 AND MAXSUPPORT = 5

Model	In-Domain Accuracy (%)					
	CUB	Textures	Fungi	VGG flower	Traffic Signs	CIFAR 100
k-NN [27]	64.1 ± 0.6	36.8 ± 0.4	45.9 ± 0.6	73.8 ± 0.5	45.9 ± 0.5	50.1 ± 0.5
Finetune [57]	57.0 ± 1.7	38.3 ± 1.2	45.8 ± 1.8	73.9 ± 1.2	50.1 ± 1.3	54.5 ± 1.4
MatchingNet [41]	49.7 ± 0.5	36.7 ± 0.4	38.2 ± 0.5	66.1 ± 0.4	52.6 ± 0.4	46.9 ± 0.4
ProtoNet [42]	44.9 ± 0.5	33.5 ± 0.3	35.0 ± 0.4	56.2 ± 0.5	35.7 ± 0.4	42.8 ± 0.5
fo-MAML [58]	59.5 ± 0.6	38.5 ± 0.4	44.0 ± 0.5	70.3 ± 0.5	49.0 ± 0.4	49.7 ± 0.5
RelationNet [43]	60.5 ± 1.9	38.8 ± 1.3	46.8 ± 1.9	72.8 ± 1.4	**79.9 ± 1.2**	50.9 ± 1.6
fo-Proto-MAML [26]	47.0 ± 1.5	33.3 ± 1.1	35.5 ± 1.4	60.1 ± 1.4	36.6 ± 1.2	41.9 ± 1.3
CNAPs [27]	**65.6 ± 0.6**	41.5 ± 0.5	46.5 ± 0.5	69.7 ± 0.9	43.2 ± 0.7	55.7 ± 0.5
SimpleCNAPs [59]	64.0 ± 0.5	44.9 ± 0.9	49.8 ± 0.5	73.1 ± 0.4	48.5 ± 0.4	**64.8 ± 0.5**
DC-DML	62.2 ± 0.5	**49.5 ± 0.5**	**50.6 ± 0.6**	**82.2 ± 0.4**	48.7 ± 0.4	59.7 ± 0.5

Model	Out-of-Domain Accuracy (%)				Average Accuracy (%)		
	NWPU-RESISC45	AID	UC	WHU-RS19	In-Domain	Out-of-Domain	Overall
k-NN [27]	42.1 ± 0.5	45.9 ± 0.5	51.8 ± 0.6	54.6 ± 0.7	52.8 ± 0.5	48.6 ± 0.6	51.1 ± 0.5
Finetune [57]	55.5 ± 1.6	60.2 ± 1.7	63.6 ± 1.7	76.1 ± 1.5	53.3 ± 1.4	63.9 ± 1.6	57.5 ± 1.5
MatchingNet [41]	46.9 ± 0.5	51.3 ± 0.5	54.2 ± 0.5	65.0 ± 0.5	48.4 ± 0.4	54.4 ± 0.5	50.8 ± 0.5
ProtoNet [42]	42.2 ± 0.4	44.6 ± 0.4	51.1 ± 0.5	54.2 ± 0.5	41.4 ± 0.4	48.0 ± 0.5	44.0 ± 0.4
fo-MAML [58]	44.5 ± 0.5	50.0 ± 0.5	52.1 ± 0.5	60.8 ± 0.5	51.8 ± 0.5	51.9 ± 0.5	51.8 ± 0.5
RelationNet [43]	49.3 ± 1.5	52.8 ± 1.6	56.6 ± 1.8	65.4 ± 1.6	58.3 ± 1.6	56.0 ± 1.4	57.4 ± 1.5
fo-Proto-MAML [26]	41.6 ± 1.4	44.2 ± 1.5	50.9 ± 1.4	52.6 ± 1.4	42.4 ± 1.3	47.3 ± 1.4	44.4 ± 1.4
CNAPs [27]	49.1 ± 0.5	54.6 ± 0.5	66.9 ± 0.6	61.8 ± 0.5	53.7 ± 0.6	58.1 ± 0.5	55.5 ± 0.6
SimpleCNAPs [59]	57.0 ± 0.5	65.2 ± 0.5	74.6 ± 0.5	74.5 ± 0.5	57.5 ± 0.5	67.8 ± 0.5	61.6 ± 0.5
DC-DML	**60.6 ± 0.5**	**68.6 ± 0.6**	**78.9 ± 0.5**	**80.9 ± 0.5**	**58.8 ± 0.5**	**72.3 ± 0.5**	**64.2 ± 0.5**

TABLE II

FEW-SHOT CLASSIFICATION ACCURACY OF DC-DML COMPARED WITH THE PREVAILING APPROACHES. MAXCLASSES = 10 AND MAXSUPPORT = 10

Model	In-Domain Accuracy (%)					
	CUB	Textures	Fungi	VGG flower	Traffic Signs	CIFAR 100
k-NN [27]	59.2 ± 1.6	36.5 ± 1.2	39.3 ± 1.6	68.1 ± 1.3	42.4 ± 1.3	45.0 ± 1.3
Finetune [57]	37.4 ± 1.1	31.9 ± 1.2	33.7 ± 1.3	56.1 ± 1.3	46.8 ± 1.1	37.6 ± 1.2
MatchingNet [41]	42.3 ± 1.5	33.5 ± 1.1	31.5 ± 1.3	58.1 ± 1.4	52.9 ± 1.3	39.7 ± 1.3
ProtoNet [42]	35.4 ± 1.2	29.1 ± 1.1	24.2 ± 1.1	42.1 ± 1.3	28.4 ± 1.0	34.8 ± 1.2
fo-MAML [58]	51.3 ± 1.5	35.6 ± 1.3	37.5 ± 1.4	61.5 ± 1.3	43.2 ± 1.2	40.3 ± 1.3
RelationNet [43]	44.9 ± 1.5	33.1 ± 1.2	35.7 ± 1.7	58.3 ± 1.5	**73.4 ± 1.3**	36.8 ± 1.4
fo-Proto-MAML [26]	37.5 ± 1.4	28.3 ± 1.0	27.0 ± 1.0	48.3 ± 1.4	28.4 ± 1.0	33.0 ± 1.2
CNAPs [27]	59.5 ± 0.5	44.4 ± 0.4	46.6 ± 0.5	70.1 ± 0.4	55.8 ± 0.5	52.8 ± 0.5
SimpleCNAPs [59]	**64.6 ± 0.5**	40.4 ± 0.4	**47.8 ± 0.6**	68.6 ± 0.4	54.3 ± 0.5	51.6 ± 0.5
DC-DML	56.3 ± 0.5	**45.7 ± 0.5**	43.1 ± 0.6	**71.9 ± 0.4**	47.5 ± 0.4	**53.7 ± 0.5**

Model	Out-of-Domain Accuracy (%)				Average Accuracy (%)		
	NWPU-RESISC45	AID	UC	WHU-RS19	In-Domain	Out-of-Domain	Overall
k-NN [27]	38.2 ± 1.4	41.7 ± 1.3	49.5 ± 1.5	52.7 ± 1.5	48.4 ± 1.4	45.5 ± 1.4	47.3 ± 1.4
Finetune [57]	41.8 ± 1.3	46.8 ± 1.4	48.5 ± 1.5	61.1 ± 1.3	40.6 ± 1.2	49.6 ± 1.4	44.2 ± 1.3
MatchingNet [41]	40.0 ± 1.4	46.0 ± 1.3	48.9 ± 1.5	58.3 ± 1.3	43.0 ± 1.3	48.3 ± 1.4	45.1 ± 1.3
ProtoNet [42]	33.9 ± 1.3	35.1 ± 1.2	42.3 ± 1.3	45.9 ± 1.3	32.3 ± 1.2	39.3 ± 1.3	35.1 ± 1.2
fo-MAML [58]	39.2 ± 1.4	44.8 ± 1.3	50.1 ± 1.4	58.2 ± 1.4	44.9 ± 1.3	48.1 ± 1.4	46.2 ± 1.4
RelationNet [43]	40.2 ± 1.4	46.0 ± 1.5	49.1 ± 1.5	59.8 ± 1.3	47.0 ± 1.4	48.8 ± 1.4	47.7 ± 1.4
fo-Proto-MAML [26]	32.2 ± 1.1	33.6 ± 1.3	39.3 ± 1.4	41.6 ± 1.5	33.8 ± 1.2	36.7 ± 1.3	34.9 ± 1.2
CNAPs [27]	44.3 ± 0.5	52.0 ± 0.5	61.7 ± 0.5	60.6 ± 0.5	**54.9 ± 0.5**	54.7 ± 0.5	54.8 ± 0.5
SimpleCNAPs [59]	43.6 ± 0.5	52.7 ± 0.5	62.1 ± 0.5	61.5 ± 0.5	54.6 ± 0.5	55.0 ± 0.5	54.7 ± 0.5
DC-DML	**49.7 ± 0.6**	**56.8 ± 0.6**	**69.4 ± 0.6**	**72.2 ± 0.5**	53.0 ± 0.5	**62.0 ± 0.6**	**56.6 ± 0.5**

Tables I–III, on in-domain datasets other than the Traffic Signs dataset, our DC-DML method either achieves or approaches the highest accuracy. Only on the Traffic Signs dataset, there is a big gap between our DC-DML method and the best method. The possible reasons are as follows: some images in the Traffic Signs dataset are not bright or even fuzzy compared with those in other datasets (see Fig. 2). Our DC-DML method may be sensitive to these problems. Because our DC-DML method mainly estimates the constraint matrices according to data distribution in the task, and then the dual constrained distance metric is designed. Therefore, the deviation of feature distribution of images with low brightness or even fuzzy

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TABLE III
FEW-SHOT CLASSIFICATION ACCURACY OF DC-DML COMPARED WITH THE PREVAILING APPROACHES. MAXCLASSES = 20 AND MAXSUPPORT = 20

Model	In-Domain Accuracy (%)					
	CUB	Textures	Fungi	VGG flower	Traffic Signs	CIFAR 100
k-NN [27]	49.7 ± 0.6	45.4 ± 0.4	32.7 ± 0.5	72.5 ± 0.4	48.5 ± 0.5	44.9 ± 0.5
Finetune [57]	33.3 ± 1.0	40.4 ± 0.7	27.5 ± 0.7	54.6 ± 1.0	47.8 ± 1.0	37.0 ± 1.0
MatchingNet [41]	38.5 ± 0.6	41.8 ± 0.4	27.4 ± 0.4	63.9 ± 0.5	58.8 ± 0.5	40.5 ± 0.5
ProtoNet [42]	30.8 ± 0.6	40.0 ± 0.4	20.1 ± 0.4	45.2 ± 0.6	31.5 ± 0.4	32.8 ± 0.6
fo-MAML [58]	45.2 ± 0.6	42.9 ± 0.4	31.4 ± 0.5	63.7 ± 0.5	48.1 ± 0.5	39.5 ± 0.6
RelationNet [43]	35.3 ± 0.6	38.0 ± 0.4	27.6 ± 0.5	65.5 ± 0.5	**85.7 ± 0.3**	34.3 ± 0.5
fo-Proto-MAML [26]	35.4 ± 1.1	39.4 ± 0.7	22.7 ± 0.7	49.8 ± 1.1	33.8 ± 1.0	34.0 ± 1.0
CNAPs [27]	54.2 ± 0.6	48.9 ± 0.4	41.5 ± 0.6	**77.0 ± 0.4**	62.7 ± 0.5	51.2 ± 0.6
SimpleCNAPs [59]	**54.9 ± 0.6**	47.8 ± 0.4	**41.7 ± 0.6**	74.2 ± 0.4	71.1 ± 0.4	48.0 ± 0.6
DC-DML	48.6 ± 0.7	**54.5 ± 0.4**	36.0 ± 0.6	76.9 ± 0.4	51.3 ± 0.4	**53.0 ± 0.6**

Model	Out-of-Domain Accuracy (%)				Average Accuracy (%)		
	NWPU-RESISC45	AID	UC	WHU-RS19	In-Domain	Out-of-Domain	Overall
k-NN [27]	35.7 ± 0.5	40.5 ± 0.5	49.2 ± 0.5	53.8 ± 0.5	49.0 ± 0.5	44.8 ± 0.5	47.3 ± 0.5
Finetune [57]	37.1 ± 1.0	41.4 ± 1.0	46.9 ± 1.0	56.1 ± 1.0	40.1 ± 0.9	45.4 ± 1.0	42.2 ± 1.0
MatchingNet [41]	35.9 ± 0.5	40.6 ± 0.5	44.7 ± 0.6	54.6 ± 0.5	45.2 ± 0.5	44.0 ± 0.5	44.7 ± 0.5
ProtoNet [42]	29.4 ± 0.6	30.5 ± 0.6	38.2 ± 0.6	41.7 ± 0.7	33.4 ± 0.5	35.0 ± 0.6	34.0 ± 0.6
fo-MAML [58]	32.4 ± 0.5	36.3 ± 0.5	42.4 ± 0.6	48.8 ± 0.6	45.1 ± 0.5	40.0 ± 0.6	43.1 ± 0.5
RelationNet [43]	31.9 ± 0.5	37.5 ± 0.6	40.9 ± 0.6	50.7 ± 0.6	47.7 ± 0.5	40.3 ± 0.6	44.7 ± 0.5
fo-Proto-MAML [26]	28.4 ± 1.0	30.7 ± 1.0	36.4 ± 1.1	40.5 ± 1.1	35.9 ± 0.9	34.0 ± 1.1	35.1 ± 1.0
CNAPs [27]	38.8 ± 0.6	44.2 ± 0.6	57.8 ± 0.5	52.5 ± 0.6	55.9 ± 0.5	48.3 ± 0.6	52.9 ± 0.5
SimpleCNAPs [59]	40.0 ± 0.6	45.5 ± 0.6	55.6 ± 0.6	50.3 ± 0.7	**56.3 ± 0.6**	47.9 ± 0.6	52.9 ± 0.6
DC-DML	**43.3 ± 0.7**	**48.6 ± 0.7**	**62.8 ± 0.6**	**63.0 ± 0.6**	53.4 ± 0.6	**54.4 ± 0.7**	**53.8 ± 0.6**

characteristics is generally large, and this deviation in our DC-DML method has a greater negative effect on the final classification than other algorithms.

2) Out-of-Domain Performance: This section analyzes the classification performance of out-of-domain, in which there are mainly four datasets, i.e., NWPU-RESISC45, AID, UC Images, and WHU-RS19. For Table I, DC-DML obtains the top-1 classification accuracies of 60.6%, 68.6%, 78.9%, and 80.9% on these four out-of-domain datasets, achieving 3.6%, 3.4%, 4.3%, and 4.8% improvements on top of the comparison approaches, respectively. The out-of-domain average accuracy of DC-DML is 73.3%, which is 5.5% higher than the best comparison method. In Table II, we obtain 5.4%, 4.1%, 7.3%, and 10.7% gains over the best comparison approaches, respectively. DC-DML also achieves 62.0% top-1 out-of-domain classification performance, which is 7% higher when compared with the suboptimal approach. From Table III, we can still see that our DC-DML consistently outperforms the current prevailing algorithms, obtaining 6.1% improvement over the best one about the out-of-domain performance. DC-DML achieves the best classification accuracy of 43.3%, 48.6%, 62.8%, and 63.0% on the four out-of-domain datasets, respectively. In summary, our DC-DML obtains the state-of-the-art classification performance on four out-of-domain aerial image datasets, which proves the effectiveness of our method.

3) Overall Performance: This is the overall average accuracy of all the datasets, both in and out of the domain, to verify the overall performance of DC-DML. First, from Table I, we can see that the top-1 average accuracy of 64.2% is achieved by our DC-DML, which exceeds the best algorithm by a 2.6% margin. In Table II, DC-DML obtains 56.6% top-1 accuracy, achieving a 1.9% improvement over the best one. Likewise, from Table III, our DC-DML achieves 53.8% classification performance, which is 0.9% higher than the best one. This means that DC-DML not only generalizes to out-of-domain aerial image datasets better than the comparison approaches but also maintains the classification performance of the in-domain, thus achieving excellent overall classification performance.

D. Ablation Analysis

To study the effectiveness of the dual constrained distance we designed, we replace it within our framework with some other distance metrics, such as absolute difference, squared Euclidean, cosine similarity, Mahalanobis distance, and negative dot-product. We do this ablation study on AIFS-DATASET, where MAXCLASSES and MAXSUPPORT are set to 5, and the experimental results are shown in Table IV. We observe that the experimental performance obtained using the Mahalanobis distance is superior to that of the other four distance metrics (i.e., absolute difference, squared Euclidean, cosine similarity, and negative dot-product), but inferior to that of our dual constrained distance. Concretely, Mahalanobis distance obtains the highest classification performances of 64.0% ± 0.5% and 64.8% ± 0.5% in the CUB-200-2011 and CIFAR 100 datasets, respectively. But on other datasets, including aerial image datasets and VGG Flower, Traffic Signs, etc., our DC-DML achieves the highest classification accuracies. The best in-domain, out-of-domain, and overall average accuracies are also obtained using our dual constrained distance metric. These ablation results show

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TABLE IV

FEW-SHOT CLASSIFICATION ACCURACY OF DC-DML COMPARED WITH OTHER DISTANCE METRICS, INCLUDING ABSOLUTE DISTANCE, SQUARED EUCLIDEAN, COSINE SIMILARITY, MAHALANOBIS DISTANCE, AND NEGATIVE DOT-PRODUCT. MAXCLASSES = 5 AND MAXSUPPORT = 5

Metric	In-Domain Accuracy (%)					
	CUB	Textures	Fungi	VGG flower	Traffic Signs	CIFAR 100
Absolute Distance	31.4 ± 0.4	39.7 ± 0.4	39.1 ± 0.5	51.1 ± 0.4	34.0 ± 0.4	36.5 ± 0.4
Squared Euclidean	32.8 ± 0.4	44.7 ± 0.4	39.3 ± 0.5	49.0 ± 0.4	32.4 ± 0.3	36.3 ± 0.4
Cosine Similarity	30.9 ± 0.4	38.4 ± 0.4	35.1 ± 0.5	50.7 ± 0.4	32.9 ± 0.4	35.0 ± 0.4
Negative Dot Product	30.2 ± 0.4	46.5 ± 0.4	31.9 ± 0.4	51.3 ± 0.5	29.2 ± 0.3	34.5 ± 0.4
Mahalanobis Distance	**64.0 ± 0.5**	44.9 ± 0.9	49.8 ± 0.5	73.1 ± 0.4	48.5 ± 0.4	**64.8 ± 0.5**
DC-DML (w/o penalty)	62.8 ± 0.5	46.6 ± 0.5	50.5 ± 0.6	78.1 ± 0.4	45.6 ± 0.4	60.6 ± 0.5
DC-DML (w/ penalty)	62.2 ± 0.5	**49.5 ± 0.5**	**50.6 ± 0.6**	**82.2 ± 0.4**	**48.7 ± 0.4**	59.7 ± 0.5

Metric	Out-of-Domain Accuracy (%)				Average Accuracy (%)		
	NWPU-RESISC45	AID	UC	WHU-RS19	In-Domain	Out-of-Domain	Overall
Absolute Distance	44.5 ± 0.5	63.5 ± 0.6	62.7 ± 0.6	69.2 ± 0.5	38.6 ± 0.4	60.0 ± 0.6	47.2 ± 0.5
Squared Euclidean	43.5 ± 0.5	62.5 ± 0.6	60.4 ± 0.6	68.1 ± 0.5	39.1 ± 0.4	58.6 ± 0.6	46.9 ± 0.5
Cosine Similarity	37.8 ± 0.4	51.5 ± 0.5	58.4 ± 0.5	58.4 ± 0.5	37.2 ± 0.4	51.5 ± 0.5	42.9 ± 0.4
Negative Dot Product	35.7 ± 0.4	47.5 ± 0.6	51.6 ± 0.6	56.6 ± 0.6	37.3 ± 0.4	47.9 ± 0.6	41.5 ± 0.5
Mahalanobis Distance	57.0 ± 0.5	65.2 ± 0.5	74.6 ± 0.5	74.5 ± 0.5	57.5 ± 0.5	67.8 ± 0.5	61.6 ± 0.5
DC-DML (w/o penalty)	58.4 ± 0.5	**68.8 ± 0.5**	76.0 ± 0.5	78.5 ± 0.5	57.4 ± 0.5	70.4 ± 0.5	62.6 ± 0.5
DC-DML (w/ penalty)	**60.6 ± 0.5**	68.6 ± 0.6	**78.9 ± 0.5**	**80.9 ± 0.5**	**58.8 ± 0.5**	**72.3 ± 0.5**	**64.2 ± 0.5**

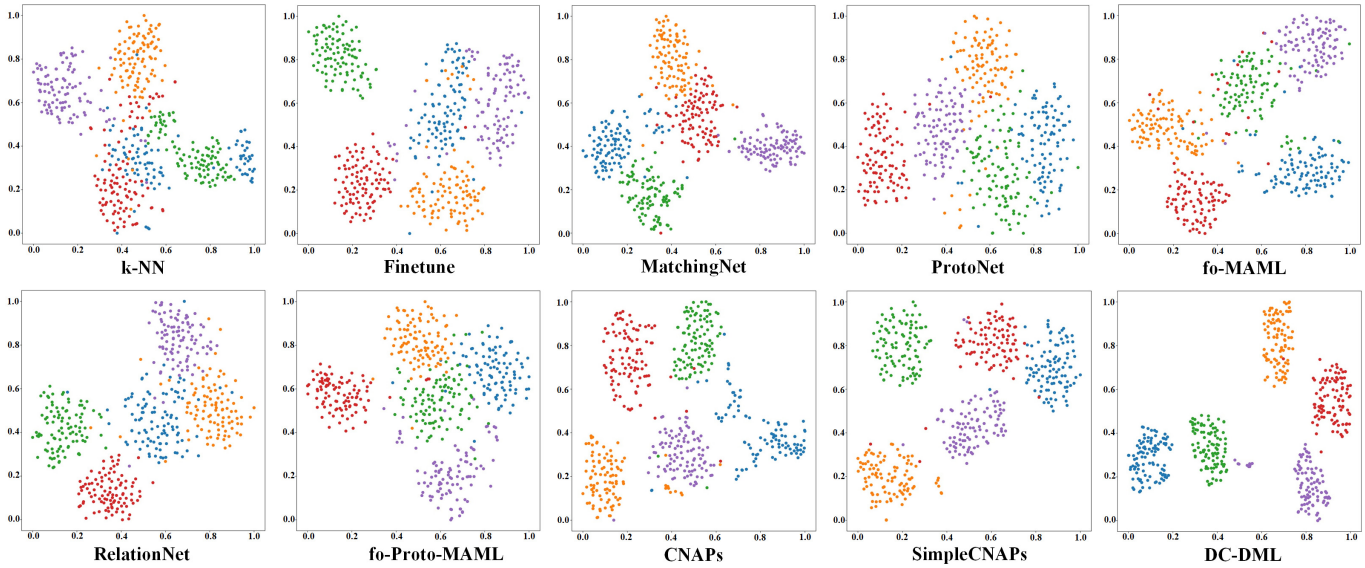


Fig. 5. t-SNE visualization of all the compared methods. Hundred images are randomly sampled from five random unseen classes on the AID dataset.

that our dual constrained distance improves the classification performance over other commonly used distance metrics on AIFS-DATASET.

In addition, we add a cross-class bias penalty into the objective loss function to further improve the classification performance. To verify the contribution of this cross-class bias penalty of our method, we compare "DC-DML (w/o penalty)," where the cross-class bias penalty is not included in the objective function, with "DC-DML (w/ penalty)" that uses the cross-class bias penalty. From Table IV, while removing the cross-class bias penalty, the classification performance of "DC-DML (w/o penalty)" is slightly lower than that of "DC-DML (w/ penalty)." By attaching this cross-class bias penalty to the objective function to train and optimize the FSL model, we obtain 1.4%, 1.9%, and 1.6% improvements, respectively, for the in-domain, out-of-domain, and overall average performance of AIFS-DATASET. Even without the cross-class bias penalty term, "DC-DML (w/o penalty)" is still superior to other distance metrics on the whole. Inspired by [60], as our work only considers the issues of inter-class similarity and intra-class diversity in aerial images, we intend to overcome the problems of various variabilities (e.g., noise effects) in our future work.

In order to more intuitively illustrate the effectiveness of our DC-DML, we carry out t-SNE visualization of the test sample embeddings of DC-DML. Fig. 5 shows the t-SNE visualization of all the compared methods. From Fig. 5, it can be observed that problems such as high dispersion of intra-class sample embedding and fuzzy class boundary exist in the compared methods, while our DC-DML not only does not have these problems but also can reach a good feature discrimination level for unseen classes.

We also compare the FLOPs and fps of our DC-DML with other methods on NWPU-RESISC45, and we summarize the comparisons in Table V. Our DC-DML is derived from CNAPs; from Table V, we can see that our method outperforms

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TABLE V
COMPARISONS OF FLOPS AND fps ON NWPU-RESISC45. MAXCLASSES = 20 AND MAXSUPPORT = 20

	FLOPs	fps
k-NN [27]	13.64	432.51
Finetune [57]	89.42	17.52
MatchingNet [41]	1.58	443.95
ProtoNet [42]	56.44	430.40
fo-MAML [58]	79.32	152.98
RelationNet [43]	3.13	391.15
fo-Proto-MAML [26]	79.29	206.77
CNAPs [27]	108.86	147.48
SimpleCNAPs [59]	107.99	141.19
DC-DML	107.96	141.98

CNAPs by 4.5% with less FLOPs and pretty much fps. For other algorithms, although our method is slightly worse than these algorithms in terms of FLOPs and fps, our DC-DML improves the accuracy by 3.3%–14.9%.

V. CONCLUSION

In this work, we propose a new and more realistic benchmark, termed as AIFS-DATASET, to rigorously train the FSL model and test the generalization ability of the model in aerial image scene classification. This AIFS-DATASET contains several diverse datasets and can generate heterogeneous meta-tasks. In addition, we also designed a new metric learning method for this AIFS-DATASET, called Dual Constrained Distance Metric Learning (DC-DML). DC-DML is proposed because of high requirements of AIFS-DATASET on the adaptive ability of the model, as well as the issues of inter-class similarity and intra-class diversity in aerial images. DC-DML mainly includes a task-adapted feature extractor same as CNAPs and a dual constrained distance metric. We also introduce a cross-class bias penalty into the objective function to further improve the classification performance of DC-DML. The results of comparison experiments and ablation experiments on AIFS-DATASET all prove that our algorithm is effective.

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